
MAE Preliminary Examination

Mathematics Section

Monday, April 20, 2026, 1:00pm-3:30pm

Your Name

THREE OF THE FOUR PROBLEMS WILL BE GRADED Select the 3 problems you have worked on, to be graded:		points
Problem:		/10
Problem:		/10
Problem:		/10
Total		/30

Please give your answers/work in the space provided

Explain your work/steps clearly

Linear Algebra: Problem 1 [10 points]:

Let $A, B \in \mathbb{R}^{n \times n}$ and assume that the characteristic polynomial of A is $\det(sI - A) = s^3 + s^2 + s + 1$. (Hint: one eigenvalue is -1 .)

- (a) Express A^{-1} as a polynomial function of A .
- (b) Can A be a symmetric matrix? If not, explain why not, and if yes, produce an example.
- (c) Determine the Jordan form of A ?
- (d) Explain why A is similar to the matrix

$$B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -1 & -1 \end{bmatrix}?$$

Workspace for Problem 1: Explain your reasoning/work here.

Solution

- (a) By Cayley-Hamilton, $A^3 + A^2 + A + I = 0$, hence $A^{-1} = -A^2 - A - I$.
- (b) The roots of the characteristic polynomial of A , $\{-1, i, -i\}$, includes imaginary ones. A symmetric matrix always has real eigenvalues. Hence, A cannot be symmetric.
- (c) The eigenvalues of A are as above $\{-1, \pm i\}$, hence the real Jordan form is

$$J_{A, \mathbb{R}} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

and the complex, is

$$J_{A, \mathbb{C}} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & -i \end{bmatrix}.$$

Either solution is acceptable for this problem.

- (d) This is always the case, since they have the same characteristic polynomial and the roots are simple. That is, they are both diagonalizable and share the same Jordan form over $\mathbb{C}$.

Workspace for Problem 1: Explain your reasoning/work here.

Differential Equations: Problem 2 [10 points]:

For the stationary nonlinear system

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_1 - x_2^2 \\ x_2 - x_1x_2 \end{bmatrix}$$

Do the following:

1. Find all equilibrium states, $x_{eq,i}$.
2. Determine the linearized dynamics.
3. Determine the eigenvalues of each linearized system, and classify the equilibria.
4. Sketch the phase portrait.

Workspace for Problem 2: Explain your reasoning/work here.

Solution:

1. Equilibria satisfy

$$x_1 - x_2^2 = 0, \quad x_2 - x_1x_2 = x_2 - x_2^3 = 0.$$

Thus

$$x_2 \in \{0, \pm 1\} \text{ corresponding to } x_1 = x_2^2 \in \{0, 1\}.$$

Therefore the equilibrium points are

$$(0, 0), \quad (1, 1), \quad (1, -1).$$

2. The Jacobian matrix is

$$J(x_1, x_2) = \begin{pmatrix} \frac{\partial}{\partial x_1}(x_1 - x_2^2) & \frac{\partial}{\partial x_2}(x_1 - x_2^2) \\ \frac{\partial}{\partial x_1}(x_2 - x_1x_2) & \frac{\partial}{\partial x_2}(x_2 - x_1x_2) \end{pmatrix} = \begin{pmatrix} 1 & -2x_2 \\ -x_2 & 1 - x_1 \end{pmatrix}.$$

3. Linearized dynamics & classification:

At (0, 0):

$$J(0, 0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The eigenvalues are

$$\lambda_1 = \lambda_2 = 1.$$

Since both eigenvalues are positive, (0, 0) is an **unstable node** (a source).

At (1, 1):

$$J(1, 1) = \begin{pmatrix} 1 & -2 \\ -1 & 0 \end{pmatrix}.$$

The characteristic polynomial is

$$\det \begin{pmatrix} 1 - \lambda & -2 \\ -1 & -\lambda \end{pmatrix} = (1 - \lambda)(-\lambda) - 2 = \lambda^2 - \lambda - 2.$$

Hence

$$\lambda^2 - \lambda - 2 = (\lambda - 2)(\lambda + 1) = 0,$$

so the eigenvalues are

$$\lambda_1 = 2, \quad \lambda_2 = -1.$$

Therefore (1, 1) is a **saddle point**.

At $(1, -1)$:

$$J(1, -1) = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}.$$

The characteristic polynomial is

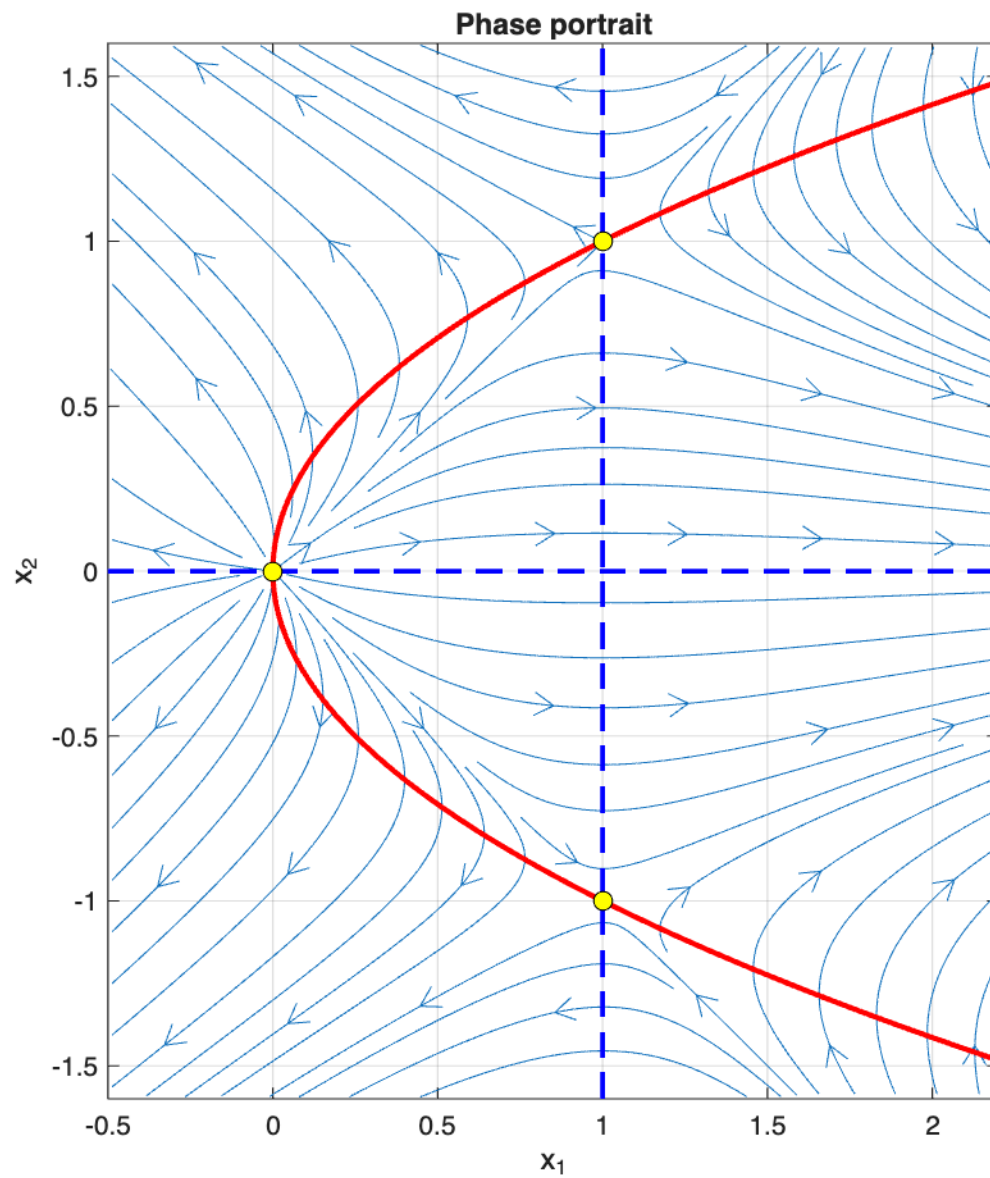
$$\det \begin{pmatrix} 1 - \lambda & 2 \\ 1 & -\lambda \end{pmatrix} = (1 - \lambda)(-\lambda) - 2 = \lambda^2 - \lambda - 2.$$

Thus the eigenvalues are again

$$\lambda_1 = 2, \quad \lambda_2 = -1.$$

Therefore $(1, -1)$ is also a **saddle point**.

4. The phase portrait is as follows:



The curve where $\dot{x}_1 = 0$:

$$x_1 - x_2^2 = 0 \quad \Longleftrightarrow \quad x_1 = x_2^2,$$

is a parabola drawn in red. The curves where $\dot{x}_2 = 0$:

$$x_2(1 - x_1) = 0 \iff x_2 = 0 \text{ or } x_1 = 1,$$

are dotted in blue.

To determine the flow direction:

- Since

$$\dot{x}_1 = x_1 - x_2^2,$$

we have

$$\dot{x}_1 > 0 \text{ when } x_1 > x_2^2, \quad \dot{x}_1 < 0 \text{ when } x_1 < x_2^2.$$

So the flow points to the right on the right side of the parabola, and to the left on the left side.

- Since

$$\dot{x}_2 = x_2(1 - x_1),$$

we have:

$$\dot{x}_2 > 0 \text{ if } \begin{cases} x_2 > 0, x_1 < 1, \\ x_2 < 0, x_1 > 1, \end{cases}$$

and

$$\dot{x}_2 < 0 \text{ if } \begin{cases} x_2 > 0, x_1 > 1, \\ x_2 < 0, x_1 < 1. \end{cases}$$

Thus the flow points upward in the upper-left and lower-right regions relative to the line $x_1 = 1$, and downward in the upper-right and lower-left regions.

In summary:

- $(0, 0)$ is a source,
- $(1, 1)$ and $(1, -1)$ are saddles.

Workspace for Problem 2: Explain your reasoning/work here.

Partial Differential Equations: Problem 3 [10 points]:

Consider Poisson's equation in the infinite strip $0 \leq y \leq d$, $-\infty < x < \infty$:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \delta(x - x_0) \delta(y - y_0),$$

with the boundary conditions

$$u(x, 0) = 0, \quad \frac{\partial u}{\partial y}(x, d) = 0,$$

where (x_0, y_0) is a point inside the strip, $0 < y_0 < d$.

Using the method of images, construct the Green's function $G(x, y; x_0, y_0)$ as an infinite series of image sources. Explain why a finite number of images cannot satisfy both boundary conditions, provide a sketch of the image system showing the locations and signs of the images, and verify that G satisfies the conditions on both walls. Comment on the convergence of the infinite series.

You may use the identity

$$\sum_{n=-\infty}^{+\infty} \ln[\alpha^2 + (\beta - nd)^2] = \ln \left[\cosh\left(\frac{\pi\alpha}{d}\right) - \cos\left(\frac{\pi\beta}{d}\right) \right] + \text{const.}$$

to express the result in closed form.

Workspace for Problem 3: Explain your reasoning/work here.

Solution

Free-space Green's function. The Green's function for the 2D Laplacian on the full plane satisfies $\nabla^2 G_\infty = \delta(\mathbf{x} - \boldsymbol{\xi})$ and is

$$G_\infty(\mathbf{x}; \boldsymbol{\xi}) = \frac{1}{2\pi} \ln r, \quad r = |\mathbf{x} - \boldsymbol{\xi}| = \sqrt{(x - \xi_1)^2 + (y - \xi_2)^2}.$$

Key insight: mixed BCs change the image pattern.

The boundary conditions are: Dirichlet ($G = 0$) on $y = 0$ and Neumann ($\partial G / \partial y = 0$) on $y = d$.

- Reflection across a *Dirichlet* wall ($y = 0$): the image has *opposite* sign.
- Reflection across a *Neumann* wall ($y = d$): the image has the *same* sign.

This is the crucial difference from the Dirichlet–Dirichlet case. The sign pattern of the images is no longer a simple alternation $+, -, +, -, \dots$

Building the image system.

Start with the source at (x_0, y_0) with sign $+1$.

1. Reflect across $y = 0$ (Dirichlet): image at $(x_0, -y_0)$ with sign -1 .
2. Reflect both across $y = d$ (Neumann, same sign): source at $y_0 \rightarrow$ image at $2d - y_0$, sign $+1$; image at $-y_0 \rightarrow$ image at $2d + y_0$, sign -1 .
3. Reflect those two new images across $y = 0$ (Dirichlet, flip sign): image at $2d - y_0 \rightarrow$ image at $-(2d - y_0)$, sign -1 ; image at $2d + y_0 \rightarrow$ image at $-(2d + y_0)$, sign $+1$.
4. Reflect across $y = d$ (Neumann, same sign): continues the pattern.

The period of the image system is $4d$ (not $2d$ as in the Dirichlet–Dirichlet case). The full set of images:

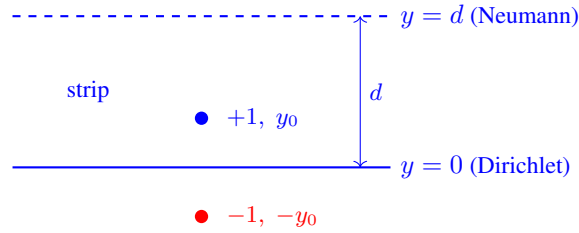
Position	Sign	Role	
$y_0 + 4nd$	$+1$	sources	$n \in \mathbb{Z}$
$-y_0 + 4nd$	-1	Dirichlet reflections of sources	
$2d - y_0 + 4nd$	$+1$	Neumann reflections of sources	
$2d + y_0 + 4nd$	-1	Dirichlet reflections of Neumann images	

• $+1, 4d+y_0$

• $-1, 4d-y_0$

• $-1, 2d+y_0$

• $+1, 2d-y_0$



• $-1, -(2d-y_0)$

• $+1, -(2d+y_0)$

The sign pattern within one period is: $+1, -1, +1, -1$ (at positions $y_0, -y_0, 2d - y_0, 2d + y_0$), repeating with period $4d$.

Why a finite number of images fails.

A single image at $-y_0$ enforces $G = 0$ on $y = 0$, but its Neumann reflection across $y = d$ creates a new source that violates $G = 0$ on $y = 0$. Each correction for one wall propagates a violation to the other. Only the infinite series achieves both conditions simultaneously.

Green's function (series form).

Grouping images by their y -positions:

$$G = \frac{1}{4\pi} \sum_{n=-\infty}^{+\infty} \left[\ln\left((x-x_0)^2 + (y-y_0-4nd)^2\right) - \ln\left((x-x_0)^2 + (y+y_0-4nd)^2\right) \right. \\ \left. + \ln\left((x-x_0)^2 + (y-(2d-y_0)-4nd)^2\right) - \ln\left((x-x_0)^2 + (y-(2d+y_0)-4nd)^2\right) \right]. \quad (1)$$

Verification on $y = 0$ (Dirichlet):

Setting $y = 0$, the first and second $\ln$ terms become $\ln[\dots + (y_0 + 4nd)^2]$ and $\ln[\dots + (y_0 - 4nd)^2]$; pairing $+n$ with $-n$ shows they cancel. Similarly, the third and fourth terms become $\ln[\dots + (2d - y_0 + 4nd)^2]$ and $\ln[\dots + (2d + y_0 + 4nd)^2]$; reindexing $m = -n$ in one and $m = n$ in the other shows cancellation. Hence $G(x, 0; x_0, y_0) = 0$. ✓

Verification on $y = d$ (Neumann):

We need $\partial G/\partial y = 0$ at $y = d$. Differentiating each $\ln$ term produces expressions of the form $\frac{2(y-Y_n)}{(x-x_0)^2+(y-Y_n)^2}$. At $y = d$, the four image positions Y_n within one period (relative to $y = d$) are $d - y_0 - 4nd$, $d + y_0 - 4nd$, $y_0 - d - 4nd$, $-y_0 - d - 4nd$. The first and third are negatives of each other (at $n \leftrightarrow -n$), contributing derivatives that cancel in pairs due to the matching signs. Similarly for the second and fourth. Hence $\partial G/\partial y|_{y=d} = 0$. ✓

Closed-form summation.

Each of the four sums in (1) is of the form $\sum_n \ln[\alpha^2 + (\beta - 4nd)^2]$. Applying the given identity with d replaced by $4d$ (i.e., the period of the image system):

$$\begin{aligned} S_1 &= \sum_n \ln[\alpha^2 + (y - y_0 - 4nd)^2] = \ln \left[\cosh \frac{\pi\alpha}{4d} - \cos \frac{\pi(y - y_0)}{4d} \right] + C, \\ S_2 &= \sum_n \ln[\alpha^2 + (y + y_0 - 4nd)^2] = \ln \left[\cosh \frac{\pi\alpha}{4d} - \cos \frac{\pi(y + y_0)}{4d} \right] + C, \\ S_3 &= \sum_n \ln[\alpha^2 + (y - 2d + y_0 - 4nd)^2] = \ln \left[\cosh \frac{\pi\alpha}{4d} - \cos \frac{\pi(y - 2d + y_0)}{4d} \right] + C, \\ S_4 &= \sum_n \ln[\alpha^2 + (y - 2d - y_0 - 4nd)^2] = \ln \left[\cosh \frac{\pi\alpha}{4d} - \cos \frac{\pi(y - 2d - y_0)}{4d} \right] + C, \end{aligned}$$

where $\alpha = x - x_0$ and the constant C is the same for all four (depends only on α and d), so it cancels in $G = \frac{1}{4\pi}(S_1 - S_2 + S_3 - S_4)$:

$$G = \frac{1}{4\pi} \ln \left[\frac{\left(\cosh \frac{\pi\alpha}{4d} - \cos \frac{\pi(y-y_0)}{4d} \right) \left(\cosh \frac{\pi\alpha}{4d} - \cos \frac{\pi(y-2d+y_0)}{4d} \right)}{\left(\cosh \frac{\pi\alpha}{4d} - \cos \frac{\pi(y+y_0)}{4d} \right) \left(\cosh \frac{\pi\alpha}{4d} - \cos \frac{\pi(y-2d-y_0)}{4d} \right)} \right], \quad \alpha = x - x_0.$$

Comment on convergence.

Each individual $\ln r_n$ diverges as $|n| \rightarrow \infty$ (logarithmic growth), so the series of individual source contributions diverges. However, within each period the *net charge* (sum of signs) is $+1 - 1 + 1 - 1 = 0$, so the images form a charge-neutral unit cell. The potential of a neutral collection of sources decays as $1/r^2$ (dipolar) or faster at large distances, ensuring that the *grouped* sum converges. This is why the series must be written with the numerator and denominator $\ln$ terms paired together (as a ratio inside the logarithm), not as separate divergent sums. The closed-form expression confirms the convergence: each $\cosh - \cos$ factor is finite and positive.

Workspace for Problem 3: Explain your reasoning/work here.

Workspace for Problem 3: Explain your reasoning/work here.

Partial Differential Equations: Problem 4 [10 points]:

Consider the wave equation on a finite domain $0 \leq x \leq 1, t \geq 0$:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = 0, \quad u_t(x, 0) = 0,$$

with the boundary conditions

$$u(0, t) = \begin{cases} t & 0 \leq t \leq 1 \\ 1 & t > 1 \end{cases}, \quad u(1, t) = 0.$$

Using the Laplace transform in time, find $u(x, t)$. Determine the exact time t^* at which the wavefront reaches the midpoint $x = 1/2$ after reflecting from $x = 1$.

Sketch u as a function of x at a time before and after t^* .

A short table of Laplace Transforms is provided.

Some Properties of the Laplace Transform		
Operation or function	$f(t)$	$\bar{f}(s) = \int_0^\infty f(t)e^{-st} dt$
First derivative	$\frac{df}{dt}$	$s\bar{f}(s) - f(0)$
Second derivative	$\frac{d^2 f}{dt^2}$	$s^2 \bar{f}(s) - sf(0) - f'(0)$
Time scaling	$f(at)$	$\frac{1}{a} \bar{f}\left(\frac{s}{a}\right)$
Time shift	$f(t-a)H(t-a)$	$e^{-as} \bar{f}(s)$
Frequency shift	$e^{at} f(t)$	$\bar{f}(s-a)$
Dirac delta function	$\delta(t-a)$	e^{-as}
Heaviside function	$H(t-a)$	$\frac{e^{-as}}{s}$
One	1	$\frac{1}{s}$
Integer power	$t^n, n = 0, 1, 2 \dots$	$\frac{n!}{s^{n+1}}$
Exponential	e^{-at}	$\frac{1}{s+a}$
Sine	$\sin(at)$	$\frac{a}{s^2 + a^2}, \Re(s) > \Im(a) $
Cosine	$\cos(at)$	$\frac{s}{s^2 + a^2}, \Re(s) > \Im(a) $
Geometric series	$\frac{1}{1-\epsilon} = \sum_{n=0}^{\infty} \epsilon^n$	$ \epsilon < 1$

Solution

Step 1: Rewrite the BC.

$$u(0, t) = t - (t - 1) H(t - 1).$$

Check: for $t < 1$ gives t ; for $t > 1$ gives $t - (t - 1) = 1$. ✓

Step 2: Laplace transform. With zero ICs, the transformed PDE is

$$s^2 \bar{u} = c^2 \bar{u}_{xx} \implies \bar{u}_{xx} - \frac{s^2}{c^2} \bar{u} = 0.$$

General solution: $\bar{u}(x, s) = A e^{-sx/c} + B e^{+sx/c}$.

Unlike the semi-infinite case, both exponentials must be retained.

Step 3: Boundary conditions in transform space.

At $x = 0$:

$$\bar{u}(0, s) = A + B = \frac{1 - e^{-s}}{s^2}.$$

At $x = 1$:

$$\bar{u}(1, s) = A e^{-s/c} + B e^{s/c} = 0.$$

From the second equation: $A = -B e^{2s/c}$. Substituting into the first:

$$-B e^{2s/c} + B = \frac{1 - e^{-s}}{s^2} \implies B(1 - e^{2s/c}) = \frac{1 - e^{-s}}{s^2} \implies B = -\frac{1 - e^{-s}}{s^2(e^{2s/c} - 1)},$$

$$A = \frac{(1 - e^{-s}) e^{2s/c}}{s^2(e^{2s/c} - 1)} = \frac{1 - e^{-s}}{s^2(1 - e^{-2s/c})}.$$

Therefore:

$$\bar{u}(x, s) = \frac{1 - e^{-s}}{s^2} \cdot \frac{e^{-sx/c} - e^{-s(2-x)/c}}{1 - e^{-2s/c}}.$$

Step 4: Geometric series expansion.

Using $\frac{1}{1 - e^{-2s/c}} = \sum_{n=0}^{\infty} e^{-2ns/c}$ for $\text{Re}(s) > 0$:

$$\begin{aligned} \bar{u}(x, s) &= \frac{1 - e^{-s}}{s^2} \sum_{n=0}^{\infty} \left[e^{-s(x+2n)/c} - e^{-s(2-x+2n)/c} \right] \\ &= \sum_{n=0}^{\infty} \left[\frac{e^{-s\tau_n^+}}{s^2} - \frac{e^{-s\tau_n^-}}{s^2} - \frac{e^{-s(1+\tau_n^+)}}{s^2} + \frac{e^{-s(1+\tau_n^-)}}{s^2} \right], \end{aligned}$$

where we define

$$\tau_n^+ = \frac{x + 2n}{c}, \quad \tau_n^- = \frac{2 - x + 2n}{c}.$$

Step 5: Inversion.

Each term has the form e^{-as}/s^2 with $\mathcal{L}^{-1}\{e^{-as}/s^2\} = (t - a) H(t - a)$.

$$u(x, t) = \sum_{n=0}^{\infty} \left[(t - \tau_n^+) H(t - \tau_n^+) - (t - \tau_n^-) H(t - \tau_n^-) - (t - 1 - \tau_n^+) H(t - 1 - \tau_n^+) + (t - 1 - \tau_n^-) H(t - 1 - \tau_n^-) \right].$$

At any finite time, only finitely many Heaviside terms are active.

Step 6: Time of first reflection reaching the midpoint.

The wavefront leaves $x = 0$ at $t = 0$ and travels at speed c .

- It first reaches the wall at $x = 1$ at time $t = 1/c$.
- It reflects (with sign change, due to the Dirichlet condition $u(1, t) = 0$) and travels back.
- It reaches the midpoint $x = 1/2$ at time $t = 1/c + (1/2)/c = 3/(2c)$.

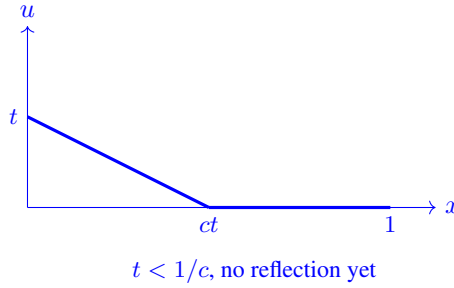
$$t^* = \frac{3}{2c}.$$

This corresponds to the activation of the $n = 0$ second term ($\tau_0^- = (2 - x)/c$) at $x = 1/2$:
 $\tau_0^- = (2 - 1/2)/c = 3/(2c) = t^*$. ✓

Step 7: Sketches.

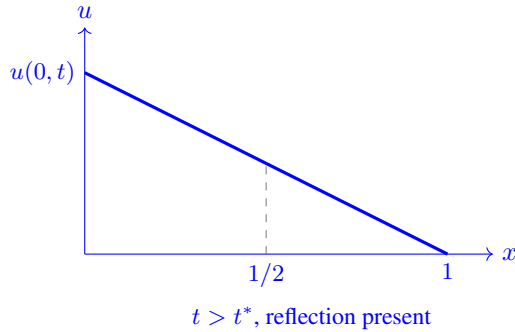
Before t^* (e.g., $t = 1/(2c)$, before any reflection):

Only the $n = 0$ first term contributes: $u = (t - x/c) H(t - x/c)$. This is a ramp from $u(0, t) = t$ down to $u = 0$ at $x = ct$.



After t^* (e.g., $t = 2/c$, after first reflection):

Two terms active for $n = 0$: the direct ramp and the reflected (negative) ramp from $x = 1$. The reflected wave subtracts from the direct wave, creating a kink.



As $t \rightarrow \infty$, the bouncing ramps accumulate and the solution approaches the steady state $u_{ss}(x) = 1 - x$ (the solution to $u_{xx} = 0$ with $u(0) = 1, u(1) = 0$).

Verification:

- $u(x, 0) = 0$: all Heavisides vanish at $t = 0$. ✓
- $u_t(x, 0) = 0$: time derivatives of each term involve Heavisides that vanish at $t = 0$ for $x > 0$. ✓
- $u(1, t) = 0$: at $x = 1$, $\tau_n^+ = (1 + 2n)/c$ and $\tau_n^- = (1 + 2n)/c$, so the terms cancel pairwise. ✓
- $u(0, t) = t - (t - 1)H(t - 1)$: at $x = 0$, $\tau_n^+ = 2n/c$ and $\tau_n^- = (2 + 2n)/c = \tau_{n+1}^+$, so consecutive terms telescope, leaving only the first: $u(0, t) = t - (t - 1)H(t - 1)$. ✓
- Each term is of the form $g(t \pm x/c)$, satisfying $u_{tt} = c^2 u_{xx}$. ✓